

Video Game Aesthetics

Rules
Geography and Representation
Number of Players
Games Without Boundaries



Heavy Rain (2010) won the hearts of critics and a number of awards for its artistic and dramatic appeal

We have already tried to reach an understanding of what games are—to give a sense of the formal qualities that we can use to decide what constitutes a game. In this chapter, we will look more closely at how to understand major game features like interactivity, rules, and **gamespace**.

Gamespace: The entire space (or world, or universe) presented by a game.

By "aesthetics" we are referring to all aspects of video games that are experienced by the player, whether directly (such as audio and graphics) or indirectly (such as rules). Thus, importantly, aesthetics as used here is not limited to how a game looks or sounds but is more broadly related to how it plays, as a function of the various design choices of the developers. Or to put it differently, Chapter 3 was about describing games as a phenomenon in contrast to other phenomena, whereas this chapter is about describing the elements that actually make up games.

- *Rules*: These defined limitations determine what you (and other characters in the game) can and cannot do, and which actions or events increase or decrease the score. In chess, for example, there is a rule that one cannot place one's own king in a position of check. In the snowboarding game *SSX 3*, there is a rule that a certain aerial maneuver gives the player a particular number of points.

- *Geography and representation*: Like the cardboard playing surface of a traditional board game, a video game's geography "physically" blocks certain actions (generally, you cannot pass through walls), while allowing others (you may be able to jump from one platform to another). The world of a video game is typically represented to the player by means of graphics and sound. Within the realm of representation there is an enormous variety of design possibilities. For instance, graphics may be two- or three-dimensional, sound may be realistic or cartoonish, and the perspective may be isometric or first-person.
- *Number of players:* In terms of design and development single-player games differ greatly from multiplayer games. In the former, computer-controlled opponents—or the environment itself—must respond entertainingly to the player's actions, while in the latter designers must ensure level playing fields, efficient communication features, and so forth.

On a very abstract level, these three elements could be independent of one another. One set of rules, for example, could be attached to a variety of game representations (we could have a game of *Star Wars* chess, which in fact we do). But, as we shall see, the choices regarding one of these elements tend in practice to shape the others.

The issue of narrative and setting is not mentioned here, since it will be dealt with separately in Chapter 7.

RULES

Recall, if you can, the 1983 blockbuster film *WarGames*. The characters played by Matthew Broderick and Ally Sheedy have come amazingly close to destroying the world—all by playing a video game. As an out-of-control military supercomputer ticks down to an imminent nuclear war, they race to reach the one man who may be able to help them, Dr. Steven Falken. Reluctant to help, Falken instead lectures them on game design. "Did you ever play tic-tac-toe?" he asks Sheedy. She answers yes but adds that she doesn't play it anymore because it is a boring game and always ends in a tie. Falken agrees: "There's no way you can win." As most of us realize at around age six, Falken is right. Unless one player makes a mistake—or is too young to grasp the structure of the game—tic-tac-toe will always end in a draw.

The savvy Broderick realizes the beauty of Falken's simple lesson, and proceeds to save the world by forcing the nuke-enabled runaway computer to play repeated games of tic-tac-toe. The goal is to teach the computer that a nuclear war between the superpowers cannot be won. In the nick of time the computer catches on and proclaims: "Strange game, the only winning move is not to play."[1]

Although deliciously dramatic, the movie's logic is hardly flawless. Strictly speaking, one does not, of course, win tic-tac-toe by not playing it. The perfect analogy would require either a cost or a disadvantage to playing the game in the first place (in the movie's plot, for example, the disadvantage

would be analogous to the cost of engaging in a nuclear war). Nor is it obvious why the computer should generalize its newfound understanding of tic-tac-toe to the Cold War as a whole, when it has played hosts of other games where one side actually can win. Nevertheless, the film demonstrates how tic-tac-toe, despite its design "flaw," is a compelling example of the functions of a game's rules.

Let's look at the incredibly simple, mundane rules of this paradigmatic nonelectronic game. In Katie Salen and Eric Zimmerman's formulation,[2] they are:

1. Play occurs on a three-by-three grid of nine empty squares.
2. Two players alternate marking empty squares, the first player marking Xs and the second player marking Os.
3. If one player places three of the same marks in a row, that player wins.
4. If the spaces are all filled and there is no winner, the game ends in a draw.

Most of us would agree that these are indeed the rules of tic-tac-toe. However, these rules alone do not guide the way we play the game. Stephen Sniderman reminds us, for example, that in standard tic-tac-toe rules, as stated above, "nothing has been said about time. Is there a time limit between moves? Normally, we both 'understand' that there is, and we both 'know' that our moves should be made within a 'reasonable' time, say 20 seconds."[3] If you realize that you have made a mistake and will lose on the other player's next turn, nothing in the four basic rules would seem to prevent you from actually never making your move, thus halting the game forever. But as Sniderman puts it:

> Anybody who seriously resorted to such a tactic would be considered childish or unsportsmanlike or socially undesirable and would probably not be asked to play in the future. This behavior seems to violate some fundamental but rarely stated principle of the game without any of us ever having to discuss it.[4]

The implication, as Salen and Zimmerman underline, is that it is not possible to state all the rules that apply to most nonelectronic games. Video games, however, are somewhat different in this regard. A game of *Microsoft Solitaire* on a Windows computer has fewer unstated rules than our paper-and-pencil game of tic-tac-toe, since everything taking place within that game must have been made possible (and legal) by the game code. Nevertheless, you could of course choose to avoid the embarrassment of quitting a hopeless game by cutting the power to the computer.

What, then, is the relationship between a game and its rules?

In a very important sense, a game *is* its rules.[5] Rules, arguably, are the most defining characteristic of games; they are the element shared by everything we usually understand as a game, and are the element that sets games apart from linear media such as novels or movies. (A novel, for example, has

its own geography, its own representation—text—and its own narrative and setting.) An important feature of rules is that they are not tied to one particular type of material; in other words, they are transmedial.[6] Thus, the game of chess does not rely on black and white pieces moving about on a physical game board. It is possible, as many die-hard Lucas fans know, to play chess using pieces representing Star Wars characters; it is just as possible to have real geographical features function as a board, and have people function as the pieces, or to have the whole thing taken care of by a computer. Thus, no matter how and where you choose to represent the chess conflict, as long as you follow the rules, you are still playing chess.

If we accept the limited definition that a game is its rules, then all games are transmedial. Of course, this does not mean that the rules of a game can be transferred into anything that someone has classified as a medium. You cannot play chess on or through a movie (although a movie can tell the story of chess, or can depict a specific chess match). The transmedial nature of games simply tells us that although a game cannot be played through any medium, a game is not tied to a specific medium.

Do you agree with the above concept? If not, perhaps you are envisioning a game of chess played using various types of Barbie dolls in a forest divided into grids—such a game would feel extremely different, no doubt, from a classical chess match with its elegant, almost minimalist, setup. In other words, you might feel that the context of the game is crucial to determining the experience of a game. You would be absolutely right, of course. But here we need to distinguish between the *formal* and the *phenomenological* levels of a game. As Salen and Zimmerman put it: "the formal system of a game, the game considered as a set of rules, is not the experience of the game."[7] Context is clearly important to any player, but when analyzing games we must sometimes bracket this acknowledgment in order to discuss how games are alike or different in terms of their more basic aesthetics.

A Definition of Rules

Science fiction author and game critic Orson Scott Card has said:

> Remember, gamewrights, the power and beauty of the art of gamemaking is that you and the player collaborate to create the final story. Every freedom that you can give to the player is an artistic victory. And every needless boundary in your game should feel to you like failure.[8]

Card's declaration will make immediate sense to many gamers and certainly has merit as a piece of game design advice. But if we believe that interactivity is central to the enjoyment of a game, does it follow that enjoyment is closely linked to player freedom? Not necessarily. Stephen Sniderman notes that "to play a game is to pursue that game's object while adhering (more or less) to its constraints."[9] Similarly, Salen and Zimmerman are adamant that one of the essential qualities of rules is that they limit player action.[10]

Many may instinctively think of rules as bad, or constraining, or a burden. We seek to fight against rules, to find ways around them. But we would do better to acknowledge that rules are an essential component of a game. These limits on our freedom as players are actually what give a shape and a drive to the playing of the game; they are what challenge us in the game world, they are what enable us to feel satisfaction when we win. Perhaps what we should hope for is not no rules at all but rules that limit player freedom in entertaining ways.

To be more formal, we can say that *a rule is an imperative governing the interaction of game objects and the possible outcome of this interaction*.

Types of Rules

Despite the brevity of the definition above, it can be difficult in practice to determine which aspects of games belong under the heading of rules. Helpfully, a number of authors have specified different types of rules. Game designer/scholar Gonzalo Frasca, for example, has proposed a dualism of rules: *ludus* rules relate to the conditions by which a player wins, and *paidia* rules refer to game procedures (such as "pawns move one square at a time" in chess).[11] As you've probably guessed, both terms are borrowed from Roger Caillois, who used them in a slightly different sense (see Chapter 3). For a more formal approach, we can turn to Jesper Juul's three levels of rules:[12]

1. *Game state rules*: These rules cover the basic aspects of the game state (i.e. the exact condition at any point in time of all game elements).
2. *Outcome valorization rules*: These rules define which outcomes are considered positive and which outcomes are considered negative.
3. *Information rules*: These rules determine what information the player receives during play about the game state.

As with Frasca, we find a distinction between rules that determine processes, and rules that relate to the outcome of the game. Juul's third category allows us to consider another important aspect of game systems: the amount of information that the player is given about the state of the entire game. To Juul, the player of *Pac-Man* has perfect (or complete)[13] information—he can see the entire screen and cannot be surprised by any new information; a *StarCraft II: Wings of Liberty* player, immersed in a world bursting with unexpected occurrences, is in a quite different position.

However, we must add that while the amount of information given may be clear, its significance may not be. One piece of information may support multiple interpretations of the game state. Imagine, for instance, that a player of *Age of Empires II: The Age of Kings* comes upon a nearly depleted and abandoned goldmine. What does this tell him about the game state? Technically, it tells him only that there is a nearly depleted mine at certain coordinates. But it may also lend credence to one or several additional hypotheses, such as "Player B has mined gold here but was disturbed and

is apparently too busy or too rich to mine the remaining gold." Thus, the information that a player actually understands can be quite different from the information that they receive.[14]

A rather different typology is suggested by Katie Salen and Eric Zimmerman. They too have three rule types:

1. *Operational rules*: These are what one would typically describe as the rules of a game. In a combination of Juul's "game state" and "outcome valorization" rules, they govern both a game's processes and its conditions for victory.
2. *Constitutive rules*: These are the underlying formal structures of a game that define its basic dynamics. For instance, some games (were one to disregard their presentational aspects) may be reduced to core logical/mathematical problems.
3. *Implicit rules*: These are all the unwritten rules that we take for granted when playing a game.

These three levels describe well the rules of nondigital games. For video games beyond a certain level of complexity (say, anything more advanced than *Tetris*), this typology is still useful but works less well in practice. The concept of constitutive rules, for example, is not often applicable; it is not easy nor necessarily meaningful to describe *Halo: Reach*—a game of vast complexity—in terms of underlying formal structures.

One is free to employ whichever typology is the most useful. But in writing this book, we have found it most useful to merely distinguish between two types of rules, along the lines of Frasca's proposal. The first type, what we call "interplay rules," determines the relationships and the properties of elements in a game. These correspond to the physical laws of the gamespace. They determine what can be done and, combined with player input, what happens. The second type, "evaluation rules," decide which occurrences are rewarded and which are punished. For instance, in *Super Mario Bros.*, one interplay rule states that Mario will jump to a certain height when a player presses the "A" button on their controller; one evaluation rule is that killing enemies by landing on them gives you points.

Gameplay

The term "gameplay" is often used but rarely defined. As commonly employed, it refers to the game dynamics, or more simply, "how it feels to play a game." Although this feeling is influenced by a game's audio and visual aspects, gameplay is usually considered a consequence of the game's rules rather than its "representation" (which we'll discuss in the following section). Using this basic definition, we can say that the gameplay of chess is deliberative, while the gameplay of *Burnout Crash!* is frantic and easily accessible. In line with this usual use of the term, we will define gameplay as *the game dynamics emerging from the interplay between rules and game*

geography.[15] These dynamics may be entertaining, or not, and they may be more or less predictable. Let us look at some examples. In *Super Monkey Ball*, the rules and geography combine to ensure that the game will be hectic and competitive and fast-paced. *Star Craft II: Wings of Liberty*, on the other hand, contains many potentially successful strategies, and, depending on the aggregate choices of the players involved, a game can be quite hectic or far more deliberative and defense-oriented.

In part, this variety in gameplay stems from the different conflicts staged by the two games. In the racing subgame of *Super Monkey Ball*, each player largely plays against the game system (even though it may not feel that way); the actions of the other players are mostly irrelevant, because your main concern is to get the monkey through the level as fast as possible. *Star Craft II*, on the other hand, provides a more chess-like geography for what is essentially a conflict between players.

Game Balance[16]

With very few exceptions, game designers attempt to achieve "balance" in every game. Essentially, this means that winning a game should be a function of player skill plus any element of randomness or luck that the game employs but should be unrelated to the game's initial conditions. In chess, for example, if black began the game with three more knights than white, the game would not be balanced. We should note that balance can exist both within the game itself (called "in-game balance") and between players ("player-player balance"). But sometimes the term also refers to a designer's attempt to balance various strategies and game units (such as soldier types) against each other in interesting ways. In a real-time strategy game, even if both sides had this option (so that the game was technically fair), it shouldn't always be best simply to concentrate on one particular unit type or strategy.

Staying with real-time strategy games, we can see how difficult it can be to achieve balance. Since each player can typically choose among various races or civilizations—each with different strengths and weaknesses—the developers must plan carefully and test meticulously to ensure that all potential interactions are balanced; further, these interactions must remain fair even across the different types of maps available, and regardless of whatever ingenious strategies players bring to the game. Examples of failure in balance are not difficult to find. In 1994's *Warcraft: Orcs and Humans*, for instance, the orc player producing warlock units will almost always win.[17] Similarly, in *Age of Empires II: The Age of Kings*, some civilizations were often considered too strong compared to others, and players often agreed not to choose these overpowerful civilizations (until a game patch fixed the issue).

One classic way of achieving in-game balance is the use of transitive relationships. The principle is well known to anyone familiar with the hand game rock-paper-scissors (Figure 5.1).

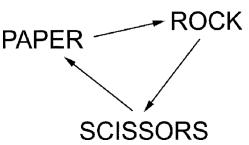


Figure 5.1 Rock-paper-scissors shows the classic principle underlying many strategy games

Rock-paper-scissors has only three strategies available to the player. But the success or failure of choosing rock, paper, or scissors depends entirely

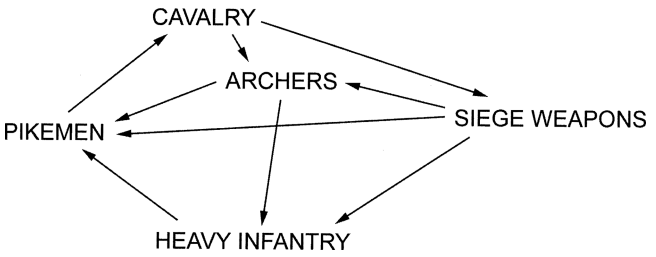


Figure 5.2 Selected unit relationships in *Age of Empires: Age of Kings* (1999). It is fundamentally the same as rock-paper-scissors but vastly more complex

on what the other player chooses; and, just as importantly, no strategy is inherently better than any of the others. For *Age of Empires II*, we can plot a simplified relationship between the game units (the types of soldiers, etc., that a player can create and use), as shown in Figure 5.2.

As we follow the arrows, we can see that the cavalry is very effective against archers and siege weapons but vulnerable to pikemen. Adding to the complexity, each of these characters (or units) has a production cost (it requires a certain amount of wood to build heavy infantry, for example, and a different amount of gold) and a production time (e.g. siege weapons take longer to build than footmen). These factors are further influenced by the specific map being used (on maps with few forests, units requiring wood are effectively more expensive to build), as well as the civilization chosen by the player (each civilization has particular advantages, such as special units). In other words, achieving in-game balance is quite complex.

When considering game balance, the field of economic game theory is helpful. Game theory is a discipline (or perhaps more accurately a set of techniques) that originates in economics, and uses the terminology of games in order to model situations of strategic interaction.[18] Of particular interest here is the concept of the Nash equilibrium (and the related concept of dominant strategies). Simply put, a game has an equilibrium state if playing a particular set of strategies produces a situation where no player can benefit from changing his strategy (assuming that the strategies used by the other player or players also remain constant). Players tend to gravitate toward this equilibrium and, once there, no one will budge (because there is no incentive to do so). In certain real-life situations this is very useful (everybody gravitates toward driving either on the left or on the right side of the road), but it is a problem for game design, because such a game will tend to be played in the same way every time. The related term "dominant strategy" refers to a strategy that is simply the best one to choose, regardless of what the other player or players are expected to do. The existence of such a strategy usually saps the game of the potential for choice, thus making it boring.

Economic game theory, while not originally intended as a tool for video game development, is highly useful for grasping the dynamics of how different rules create different player relationships. Concerned as it is with choice, it helps us (whether we are designers or analysts) understand balance and the related concept of strategy. Thus, the reader interested in these issues is encouraged to seek out (economic) game theory literature, even if it is not directly tailored toward video game analysis.